CS 24 AP Computer Science A Review

Week 2: Using Classes and Objects

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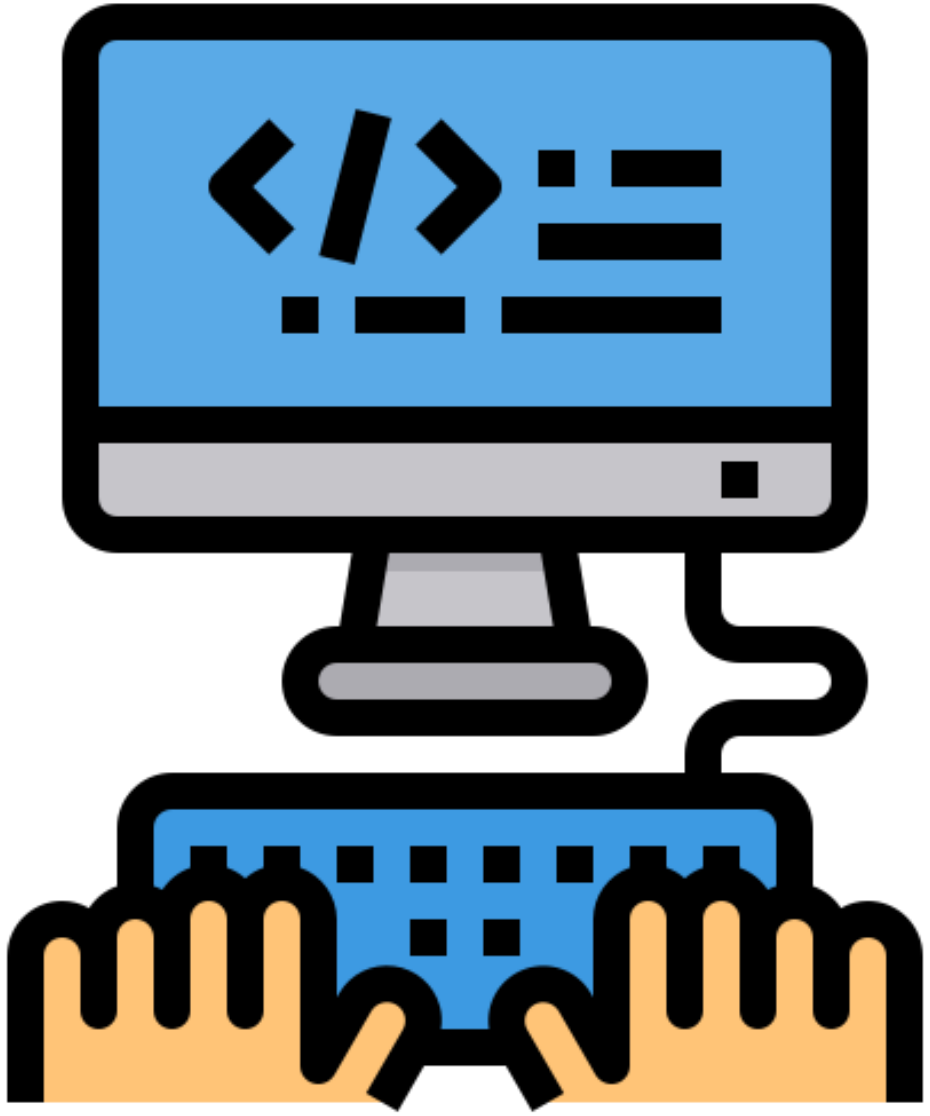
Topics

- Math Class
- Character Class
- String Class
- Immutable Objects
- Reference Variables
- Creation of Objects
- String Algorithms



Math Class

SECTION 1

Math Methods

Method	Return
Math.abs(x) // x is an int	An int equal to absolute value of x
Math.abs(x) // x is a double	A double equal to the absolute value of x
Math.pow(base, exponent)	A double equal to the base raised to the exponent
Math.sqrt(x)	A double equal to the square root of x
Math.random()	A random double in the range (0, 1)

Math Class is final

- Math class is final, Math class is also abstract
- No extension from Math Class.
- No constructor to create objects from Math
- Mostly constants and static methods

Absolute Value is the Distance

For the numbers within 3 of 5,

$$\texttt{if } (|x - 5| \leq 3)$$

Java notation,

```
if (Math.abs(x-5) <= 3)
```

Multiple Choice (1)

A math teacher is writing a program that will correctly calculate the area of a circle. Recall that the area of a circle is pi times the radius squared (πr^2). Assuming Math.PI returns an accurate decimal approximation of pi, which of the following statements WILL NOT calculate an accurate area of a circle with radius 22 ?

Multiple Choice (1)

Ⓐ `r*r*Math.PI; // r is the int 22`

Ⓑ `(double) r*r*Math.PI;`

Ⓒ `(double) (r*r)* Math.PI;`

Ⓓ All of the above choices will calculate an accurate area.

Multiple Choice (1)

Ⓐ `r*r*Math.PI; // r is the int 22`

Ⓑ `(double) r*r*Math.PI;`

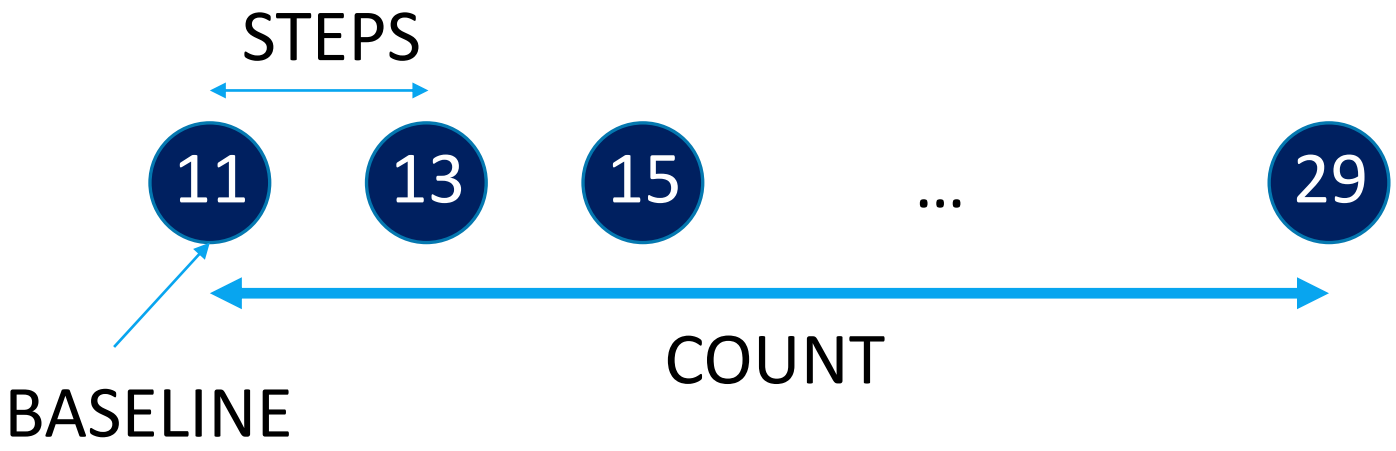
Ⓒ `(double) (r*r)* Math.PI;`

Ⓓ All of the above choices will calculate an accurate area.

Review of Random Sample Generation

int r = ((int) (Math.random()*COUNT) * STEPS + BASELINE;

int r = ((int) (Math.random() * 10) * 2 + 11);



Generation of a Real Number Range

```
double r = (Math.random()*range) + BASELINE;
```

[Baseline, Baseline+range)

[Baseline, Baseline+range]

```
double y=-999;
        while (y<-Baseline || y>Baseline+range){
            y = Math.random()*(range+0.1)-range/2;
        }
```

Block Distance

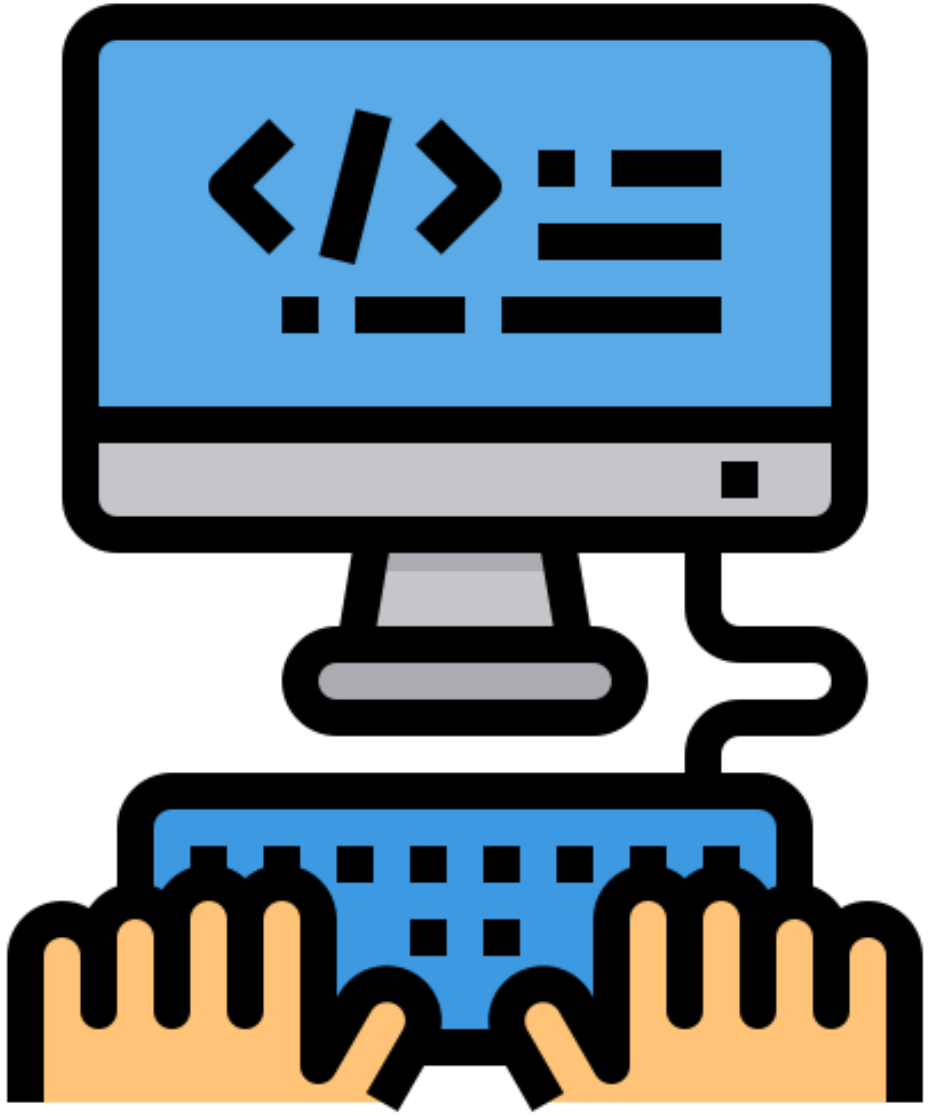
Measure the distance by how many block away from another point

D = $|x_1 - x_2| + |y_1 - y_2|$

Euclidean Distance

Euclidean distance is the "straight-line" distance between two points in Euclidean space, calculated using the Pythagorean theorem, representing the shortest path, like a bird flies. It's found by summing the squared differences of each coordinate, then taking the square root, widely used in data analysis, machine learning, and image recognition to measure similarity or difference.

$$D = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

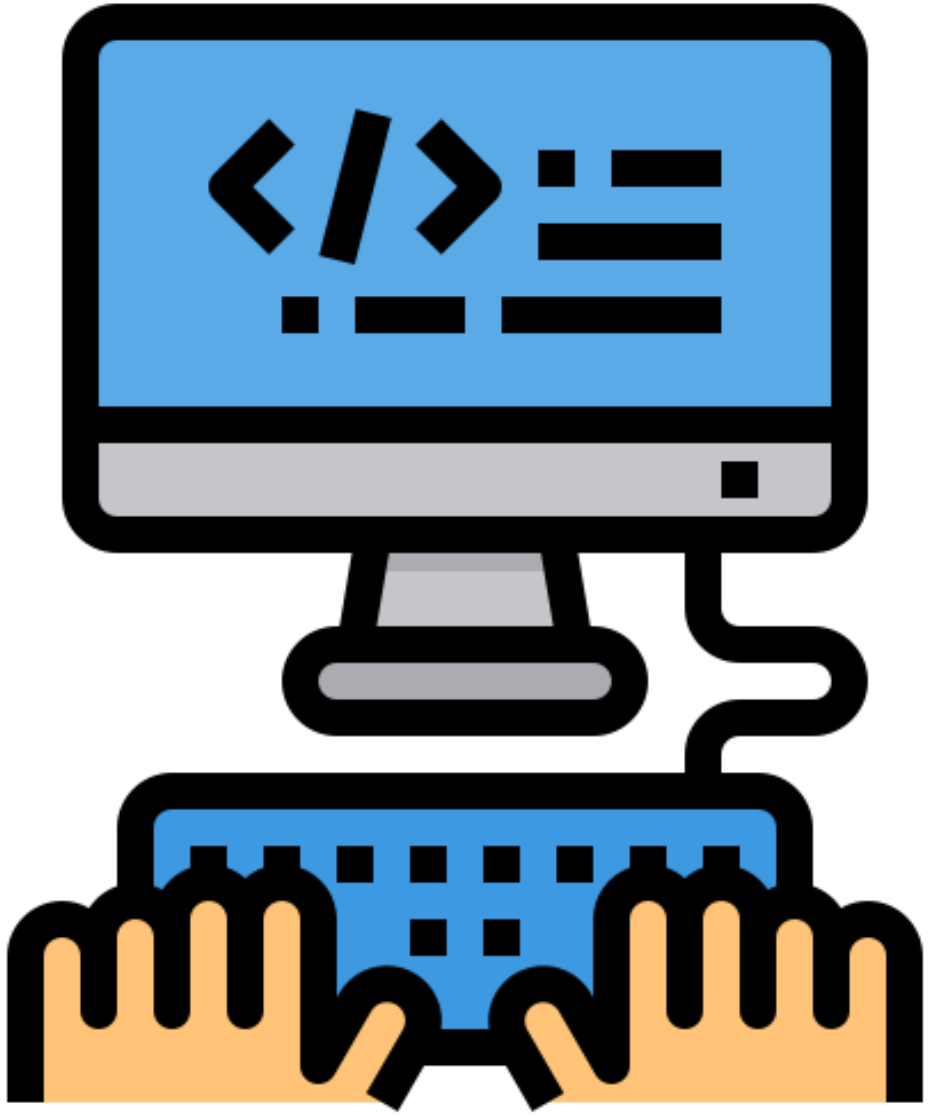


Character Class

SECTION 2

Character Methods

Method	Return
Character.isLetter(c) // c is a char	Check if c is a letter
Character.isDigit(c) // c is a char	Check if c is a digit
Chracter.toLowerCase(c) // c is a char	Convert the char c to lower case.



String Class

SECTION 3

String Methods

Method	Return
length()	Return the length of a string
charAt(i)	Return the char at index I
substring(i_s, i_e)	Return a substring from i_s to i_e (not included)
substring(i_s)	Return a substring from i_s to the end of string
equals(String s)	Check if two strings are equal
compareTo(String s)	Compare two string's alphabetical order
indexOf(String s)	Return the starting index of a pattern
indexOf(String s, from_index)	Return the starting index of a patter from index
toLowerCase()	Convert the string to lowercase
split()	Split the string into an array of strings
startsWith(String s)	Check if a string start with a pattern s

Multiple Choice (2)

Consider the following code segment:

```
String s = “This is the beginning”;
String t = s.substring(5);
int n = t.indexOf(“the”);
```

Multiple Choice (1)

Ⓐ -1

Ⓑ 3

Ⓒ 7

Ⓓ 9

Multiple Choice (1)

Ⓐ -1

❷ 3

Ⓒ 7

Ⓓ 9

Multiple Choice (3)

Consider the following code, which line will cause a compiler error?

1. String word = “Sunday Funday Celebration 1.0”;
2. word.substring(7);
3. word.substring(3, 6);
4. word.indexOf(“day”);
5. word.indexOf(1.0);

Multiple Choice (1)

Ⓐ line 2

Ⓑ line 3

Ⓒ line 4

Ⓓ line 5

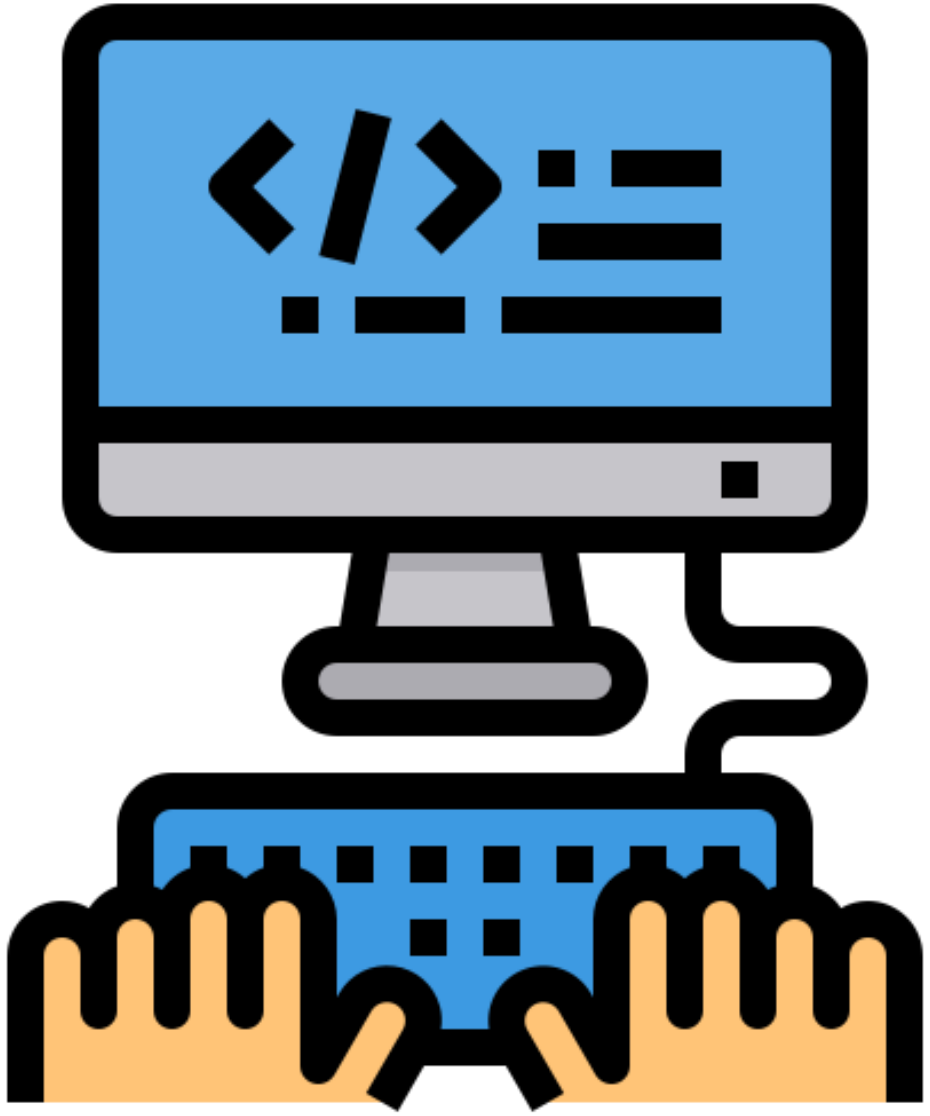
Multiple Choice (1)

Ⓐ line 2

Ⓑ line 3

Ⓒ line 4

Ⓓ line 5



Other Classes

SECTION 4

Integer Class/Double Class

```
Integer.parseInt(String s);
obj.toString();
obj.doubleValue();
Integer.valueOf(String s);
Integer I = new Integer(3);
```

Scanner Class

Creation of Input Stream:

```
Scanner input = new Scanner(new File("aa.txt"));
```

Reading Line by Line:

```
while (input.hasNext()){
      String line = input.nextLine();
 }
```

PrintWriter Class

Creating output Stream:

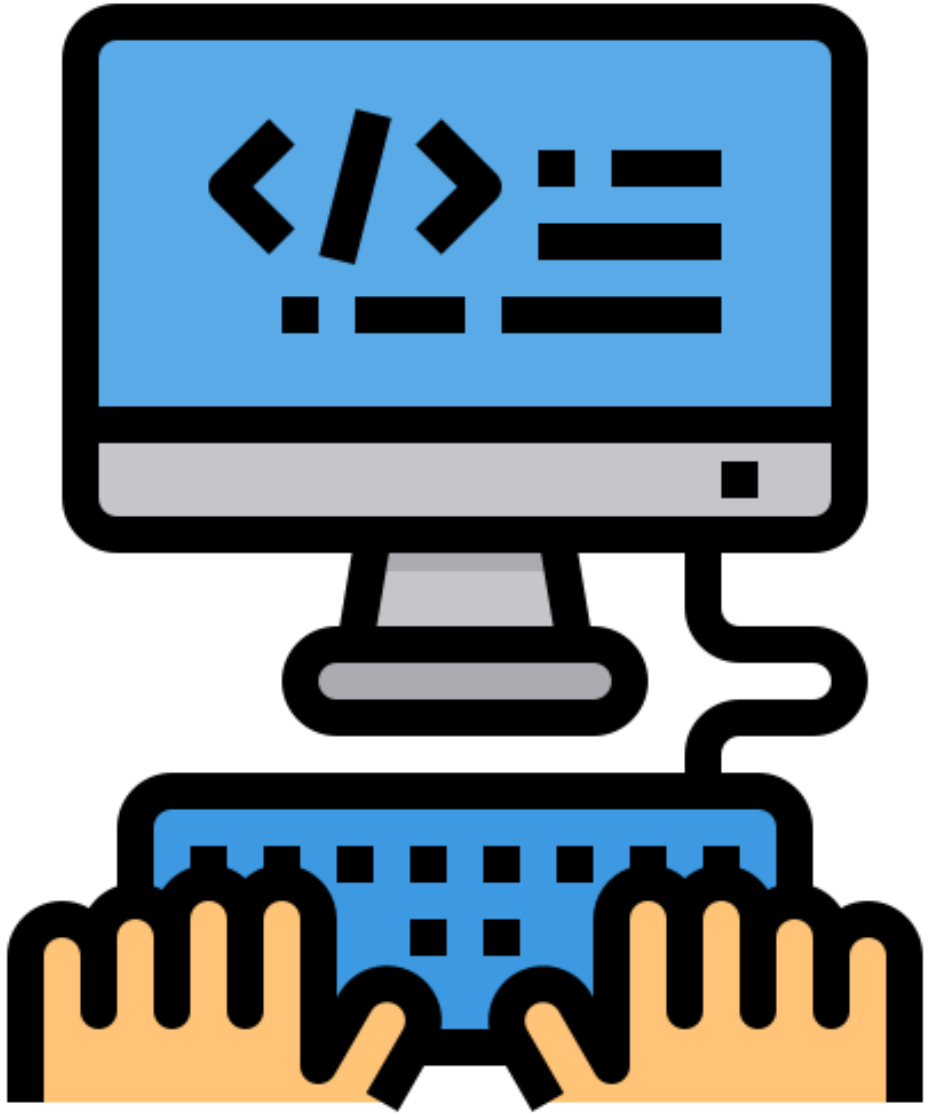
```
PrintWriter out = new PrintWriter("aa.out");
```

Output a string:

```
out.println(String s);
```

Closing the output stream:

```
out.close();
```



Immutable Objects

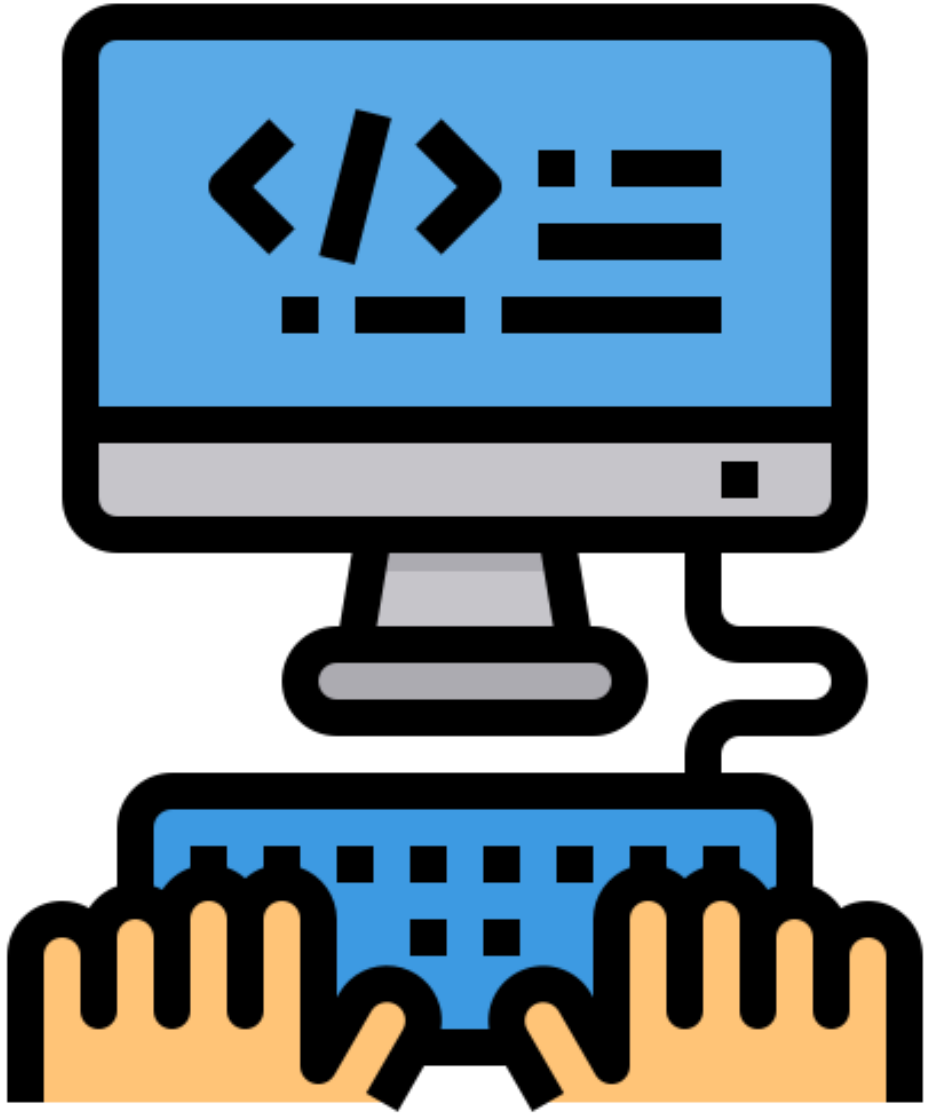
SECTION 4

Immutable Class

- All private data fields
- No mutator.
- No getter returning reference data field.

Notable Immutable Classes

- String
- Integer
- BigInteger
- BigDecimal



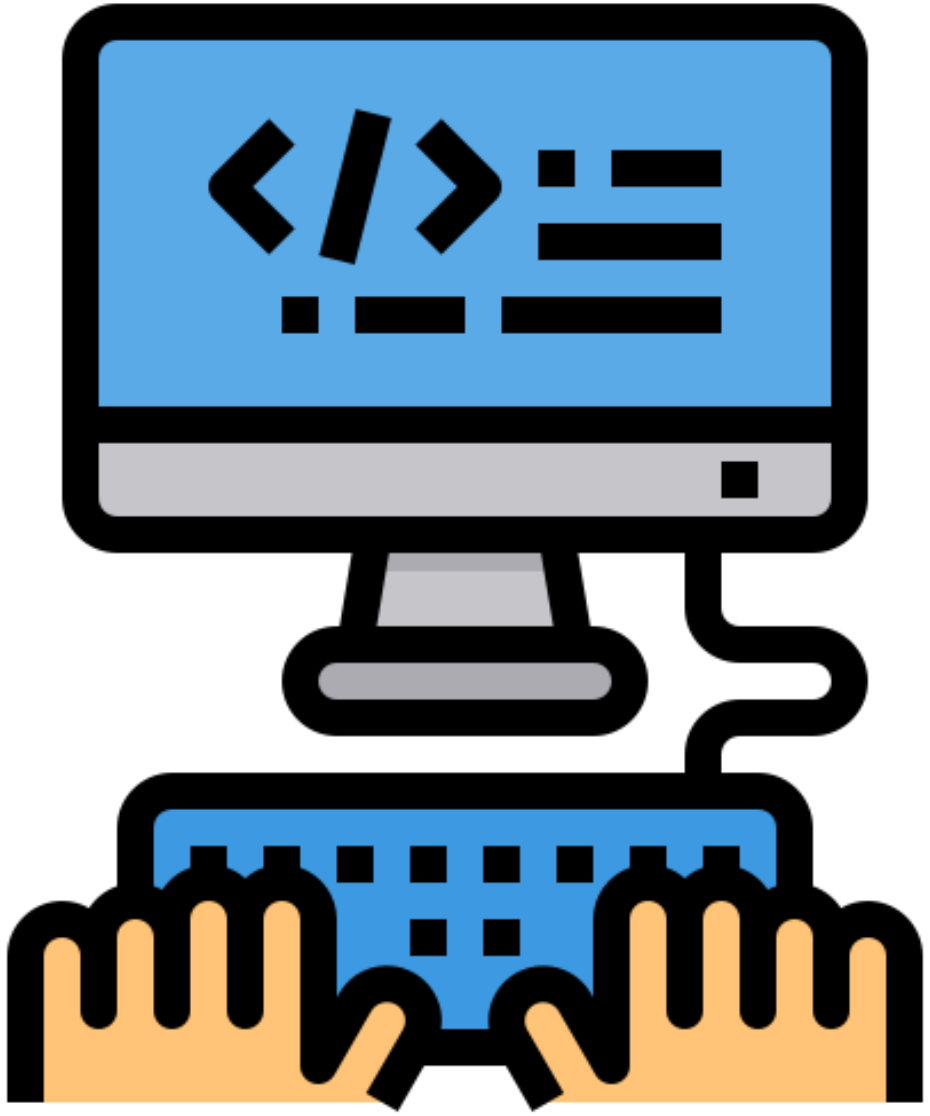
Reference Variable

SECTION 5

Reference Variable Stores the Address of the Object

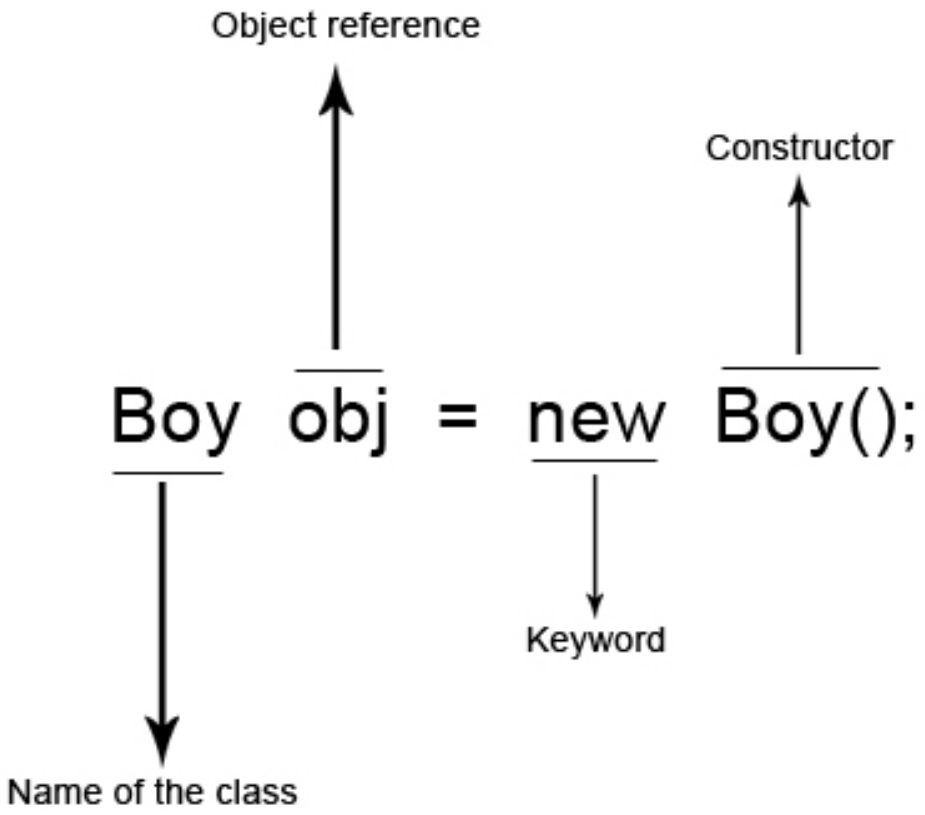
```
Car c = new Car("Honda");
```

- c is a reference variable which can be null. It can also be the reference address of an object.
- Car is the reference variable type.
- The object can be the variable type or its sub-class types.



Creation of Objects

SECTION 6



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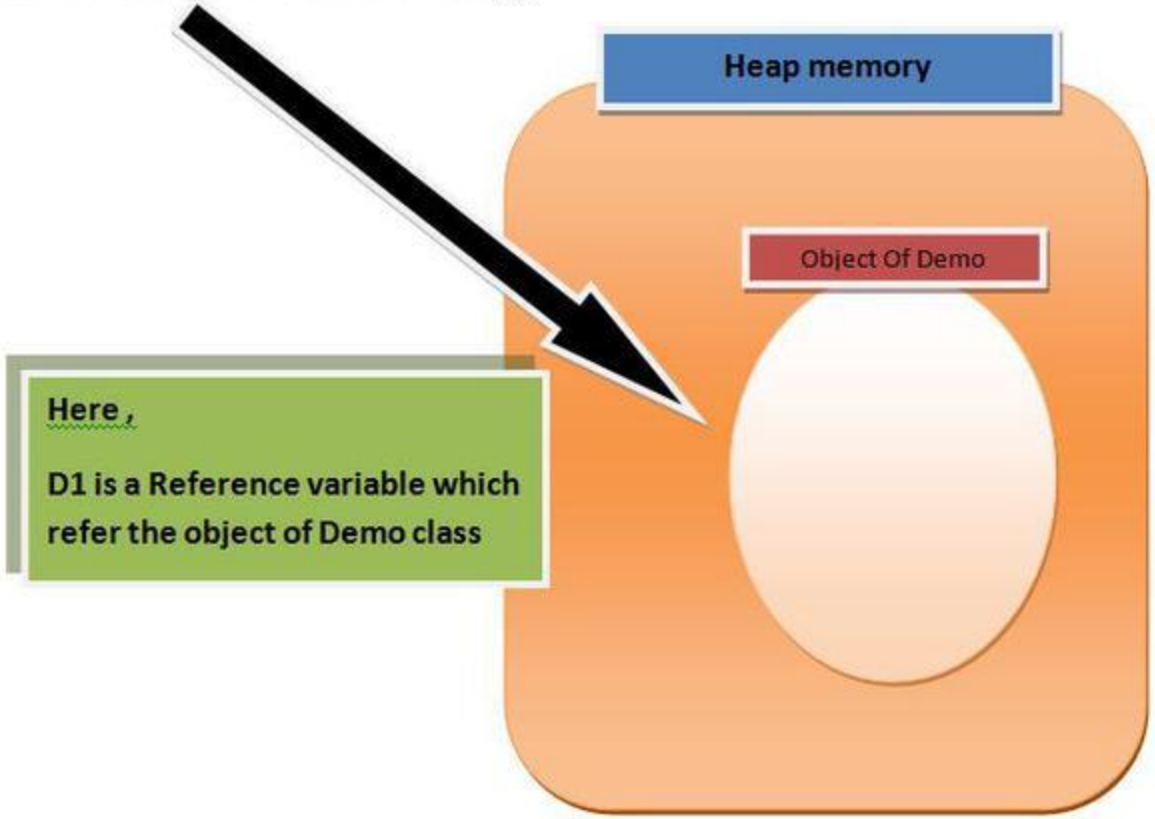
Reference Variable and Object

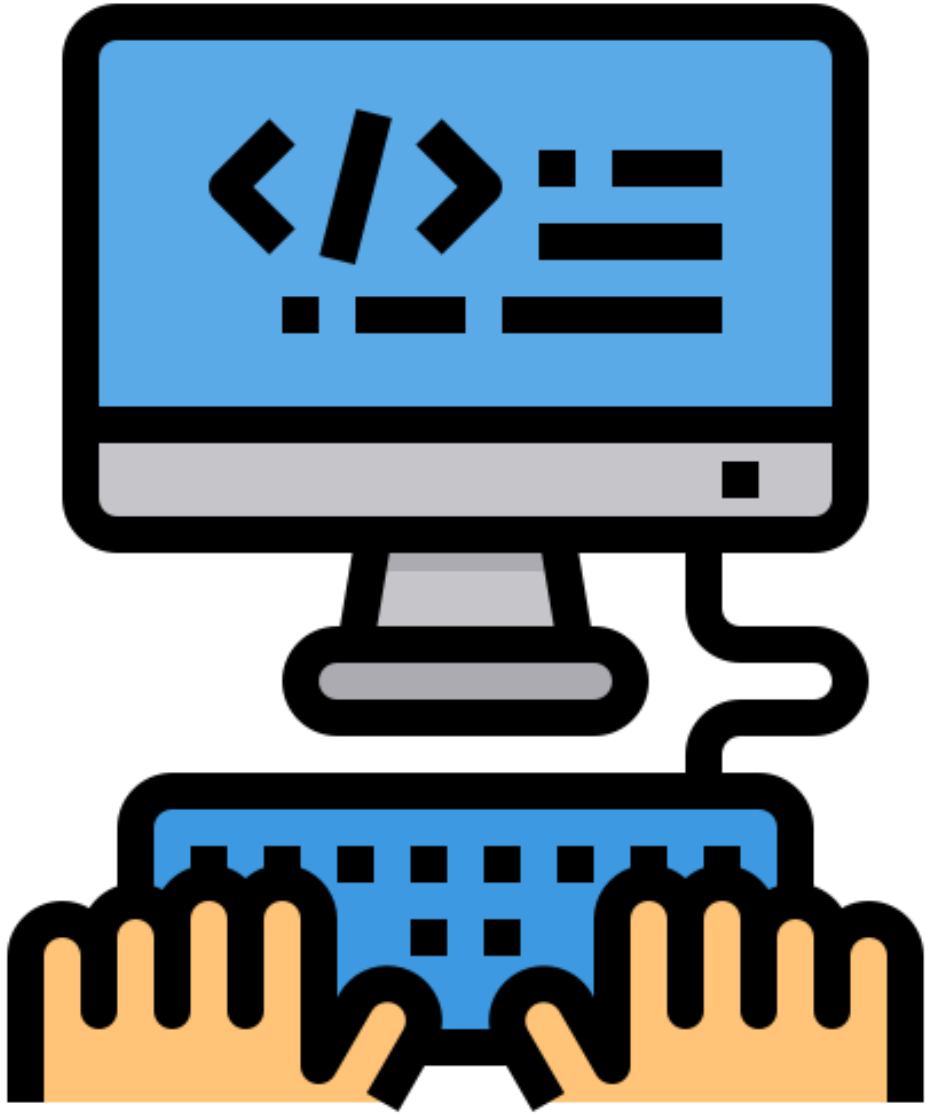
- Reference variable is a variable which can be local or global.
- The object is the data entity of the object which is stored in the dynamic memory area (Heap).

```
Demo  D1  = new Demo ();
```



Here ,

D1 is a Reference variable which refer the object of Demo class



Practice MCQ Problems

SECTION 7

Multiple Choice (1)

Given the following code, which of the following would cause a run-time error?

```
String s1 = “pizza”;
```

Ⓐ `String s2 = s1.substring(0, 2);`

Ⓑ `String s2 = s1.substring(1, 3);`

Ⓒ `String s2 = s1.substring(3, 3);`

Ⓓ `String s2 = s1.substring(4, 6);`

Multiple Choice (2)

Given the following code, which is the final value stored in s3?

```
String s1 = "kangaroo";
String s3 = s1.substring(1, 2);
```

Ⓐ "k";

Ⓑ "ka";

Ⓒ "a";

Ⓓ "an";

Multiple Choice (3)

Given the following code, which is the final value stored in s4?

```
String s1 = "tango";
String s3 = s1.substring(s1.length()-2);
```

Ⓐ "g";

Ⓑ "go";

Ⓒ "o";

Ⓓ A compiler error will occur.

Multiple Choice (4)

Given that “J” comes before “j” in the dictionary and the following variables have been defined and initialized,m which is true?

```
String str1 = “january”;
String str2 = “June”;
String str3 = “July”;
```

Ⓐ str1.compareTo(str3) < 0;

Ⓑ str3.compareTo(str3) > 0;

Ⓒ str1.compareTo(str3) < 0;

Ⓓ str1.compareTo(str2) > 0;

Multiple Choice (4)

Given the following code, what is the final value stored in p?

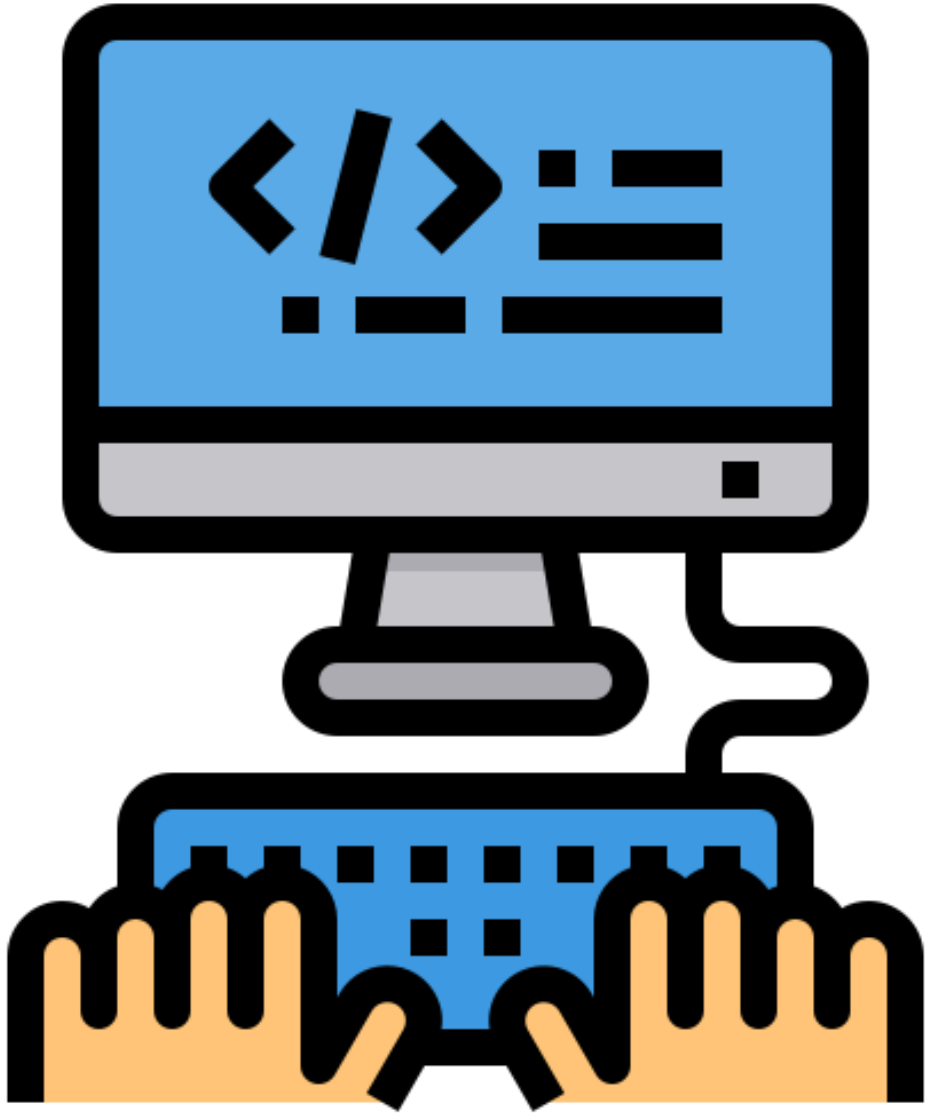
```
String word = “boottool”;
int p = word.idnexOf(“oo”);
word.substring(p+2);
P = word.indexOf(“oo”);
```

Ⓐ 0

Ⓑ 1

Ⓒ 2

Ⓓ 5



String Algorithms

SECTION 8

Reverse of Strings

```
public String reverse(String s){
    String r = “”;
    for (int i=0; i<s.length(); i++){
        r = s.charAt(i) + r;
    }
    return r:
}
```

String Pattern Searching

```
public ArrayList<Integer> find(String s, String p){
    ArrayList<Integert> plist = new ArrayList<Integer>();
    int x = s.indexOf(p);
    while (x >=0){
        plist.add(x);
        x = s.indexOf(p, x+1);
    }
    return plist;
}
```